U3
AP63205WU
L1
4.7uh
C1
100nf
C2
22nf
C8
22nf
C3
10uf
+BATT
GND
(Switch when VIN1 falls below 4V)
R1 = 30k
R2 = 10k
U4
TPS2116DRL
R1
30k
R2
10k
R3
10k
USB_5V
U5
TLV75533PDBV
C4
1uF
C5
1uF
+3V3
+5V
J2
USB_C_Receptacle_USB2.0_14P
R11
5.1k
R12
5.1k
USB_5V
D−
D+
SHIELD
SH
A1
CR1
IP4234CZ6_125
USB_DM
USB_DP
J4
Conn_01x03_Pin
U10
TLV9001IDCK
U11
TLV9001IDCK
U12
TLV9001IDCK
R13
5.1k
R14
10k
C9
3nf
R16
5.1k
R15
10k
C10
3nf
R18
5.1k
R17
10k
C11
3nf
ADC_1
ADC_2
ADC_3
Each pin has a voltage divider to 3V31. Additionaly each has a low−pass filter with a cutoff of 10kHz. This way reduction at 1kHz is only around 0.5%, while reduction at Nyquist frequency (2.105MHz) is 99.53%
XRCGB24M000F3M00R0
C7
6pF
C6
6pF
Y1
OSC_OUT
OSC_IN
U9
B3U−1000P
R4
40k
NRST
U1
STM32L433CCTx
ADC_1
ADC_2
ADC_3
TRACESWO
RXD
TXD
STBY
RGB_G
RGB_R
RGB_B
SWDIO
SWCLK
SWO
SWCLK
SWDIO
USB_DM
USB_DP
SCL
SDA
WP
J1
FTSH−105−01−L−DV−007−K
NRST
U2
FXLS8974CFR3
INT1/MOT_DET
INT2/EXT_TRIG/BOOT_OUT
INTF_SEL
SPI_CS_B/WAKE_UP
BT_MODE
SA0/SPI_MISO
SCL/SCLK
SDA/SPI_MOSI/SPI_DATA
VDD
R10
40k
R9
40k
U6
24lc32at−i/ot
D1
WL−SFTW_3528_PLCC4
R5 180
R6 300
R7 180
RGB_G
RGB_R
RGB_B
I did the resistor math by drawing 10mA from the GPIO. Idk if thats right, current kinda confuses me
U7
MCP2562−E−SN
U8
ESDCAN24−2BLY
TXD
RXD
STBY
CANH
CANL
R8
10k
J3
IEC61076−2_M8_A−coding_01x04_Receptacle_Shielded
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File: Onboarding.kicad_sch
Title:
Size: B
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1